Department of Information Technology

Academic Year: 2025-26
Semester: VI
Class / Branch / Div: TE- IT C
Subject: MAD & PWA Lab
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Date of Submission:20/03/26

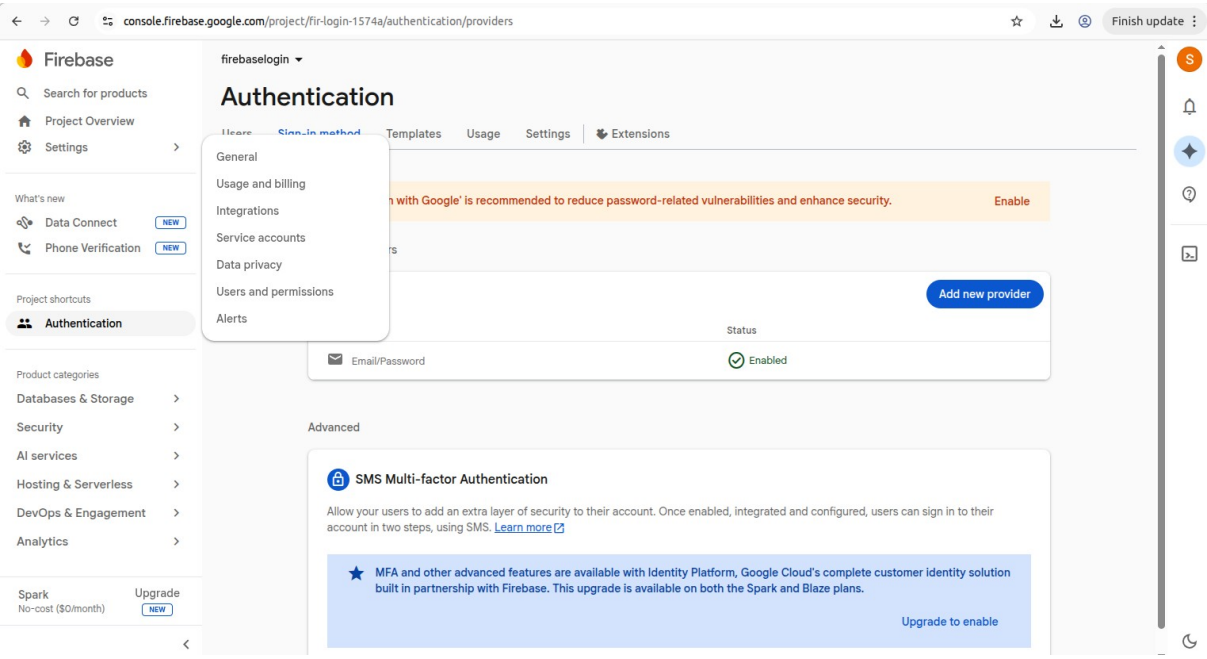
Experiment No.:7

Aim: To deploy production ready Flutter App on Android platform

Prerequisites: Android studio, flutter SDK.

Software: Android studio, flutter SDK.

Code & Output:



main.dart

```
import 'package:flutter/material.dart';
import 'package:firebase_core/firebase_core.dart';
import 'login_screen.dart';

void main() async {
```

```
  WidgetsFlutterBinding.ensureInitialized(); // Ensures Flutter framework is
initialized before Firebase starts

  await Firebase.initializeApp();   // Initialize Firebase

  runApp(MyApp());

}

class MyApp extends StatelessWidget {

  @override
  Widget build(BuildContext context) {

    return MaterialApp(
      debugShowCheckedModeBanner: false,
      title: "Firebase Login",
      home: LoginScreen(),
    );

  }

}
```

login_screen.dart

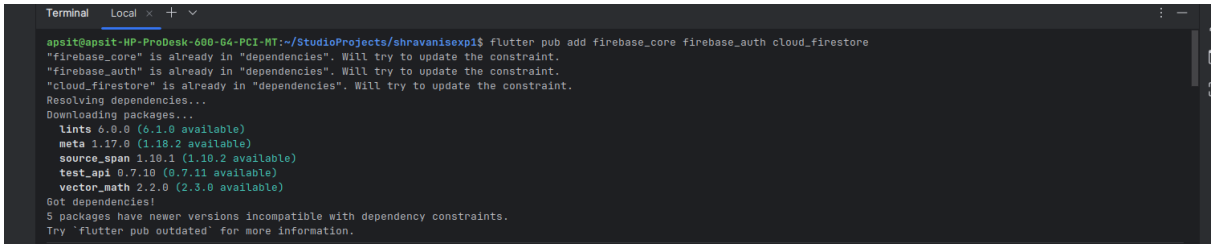
```
import 'package:flutter/material.dart';
import 'package:firebase_auth/firebase_auth.dart';

class LoginScreen extends StatefulWidget {

  @override
  _LoginScreenState createState() => _LoginScreenState();

}

class _LoginScreenState extends State<LoginScreen> {

  final TextEditingController emailController = TextEditingController();  // Controller to
```

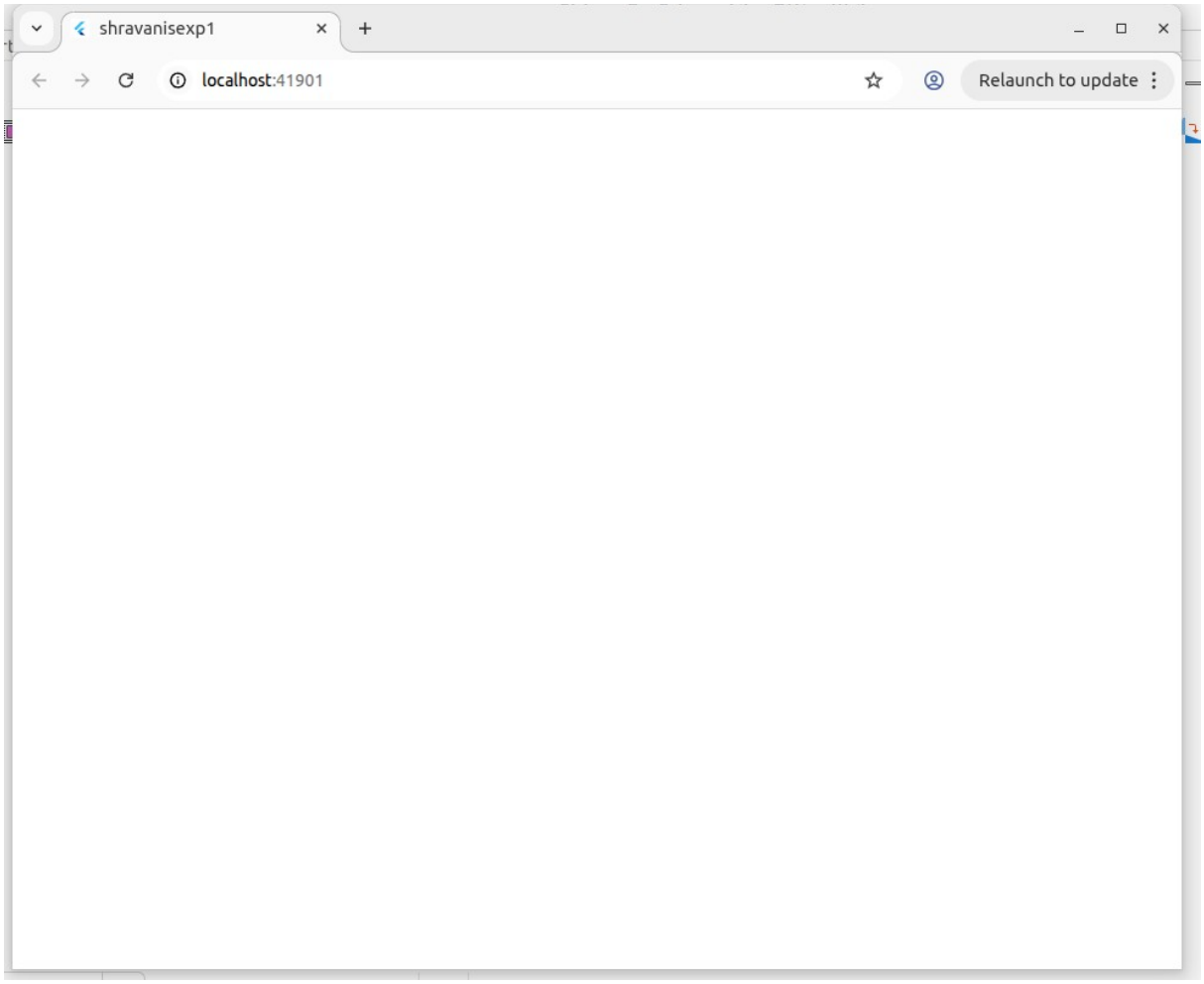
```
store email entered by user
 final TextEditingController passwordController = TextEditingController();  // Controller to
store password entered by user

 // Login function to authenticate user using Firebase
 Future loginUser() async {  // Firebase method for user login

  try {

    await FirebaseAuth.instance.signInWithEmailAndPassword(
      email: emailController.text.trim(),
      password: passwordController.text.trim(),
    );

    ScaffoldMessenger.of(context).showSnackBar(
      SnackBar(content: Text("Login Successful")),
    );

  }

  catch (e) {

    ScaffoldMessenger.of(context).showSnackBar(
      SnackBar(content: Text("Login Failed")),
    );

  }

 }

 // Function to create new user account
 Future registerUser() async {

  try {
```

```
    await FirebaseAuth.instance.createUserWithEmailAndPassword(
      email: emailController.text.trim(),
      password: passwordController.text.trim(),
    );

    ScaffoldMessenger.of(context).showSnackBar(
      SnackBar(content: Text("Registration Successful")),
    );

  }

  catch (e) {

    ScaffoldMessenger.of(context).showSnackBar(
      SnackBar(content: Text("Registration Failed")),
    );

  }

}

@override
Widget build(BuildContext context) {

  return Scaffold(

    appBar: AppBar(
      title: Text("Firebase Login"),
    ),

    body: Padding(

      padding: EdgeInsets.all(20),

      child: Column(
```

```
mainAxisAlignment: MainAxisAlignment.center,

children: [

  TextField(

    controller: emailController,

    decoration: InputDecoration(
      labelText: "Email",
      border: OutlineInputBorder(),
    ),

  ),

  SizedBox(height: 20),

  TextField(

    controller: passwordController,
    obscureText: true,

    decoration: InputDecoration(
      labelText: "Password",
      border: OutlineInputBorder(),
    ),

  ),

  SizedBox(height: 30),

  ElevatedButton(

    onPressed: () {
```

```
            loginUser();
          },
          child: Text("Login"),
        ),
        SizedBox(height: 10),
        ElevatedButton(
          onPressed: () {
            registerUser();
          },
          child: Text("Register"),
        )
      ],
    ),
  ),
);
}
}
```





Conclusion: In this experiment we have deploy flutter app on android platform